Math 372: Homework 11

Due Tuesday, April 28

Instructions: Do not write your solutions on this sheet—you don't have enough space to show your work. So use separate sheets of paper. For all problems you need to show your work.

Problem 1 (Exercise 8.1.13)**.** The calibration of a scale is to be checked by weighing a 10kg brick 25 times. Suppose that the results of different weighings are independent of one another and that the weight on each trial is normally distributed with $\sigma = .2$ kg. Let μ denote the true average weight reading on the scale.

(a) What hypotheses should be tested?

(b) With the sample mean itself as the test statistic, what is the p-value when $\overline{x} = 9.85$, and what would you conclude at significance level 0.01?

(c) For a test with $\alpha = .01$, what is the probability that recalibration is judged unnecessary when in fact $\mu = 10.1$? When $\mu = 9.8$?

Problem 2 (Exercise 8.2.19)**.** The melting point of each of 16 samples of vegetable oil was determined, resulting in $\overline{x} = 94.32$. Assume that the distribution of the melting point is normal with $\sigma = 1.20$

(a) Test $H_0 : \mu = 95$ versus $H_a : \mu \neq 95$ using a two-tailed level .01 test.

(b) If a level 0.01 test is used, what is $\beta(94)$, the probability of a type-II error when $\mu = 94$?

(c) What value of n is necessary to ensure that $\beta(94) = .1$ when $\alpha = .01$?

Problem 3 (Logistic)**.** The density function

$$f(x) = \frac{e^{3-x}}{(1 + e^{3-x})^2}$$

is an example of a *logistic* distribution.

(a) Verify that f is a pdf

(b) Find $\mathbb{P}[3 \leq X \leq 4]$

(c) Graph f. What does the mean appear to be? What about the median?

Problem 4. A balanced coin is flipped 400 times. Determine the number x such that with probability approximately 0.80, the number of heads is between $200 - x$ and $200 + x$.

Problem 5. A spinner from a board game randomly indicates a real number between 0 and 10. The spinner is fair in the sense that it indicates a number in a given interval of with the same probability as it indicates a number in any other interval of the same length.

(a) Explain why the function

$$f(x) = \begin{cases} 0.1 & : \quad 0 \leq x \leq 10 \\ 0 & : \quad \text{else} \end{cases}$$

is a probability density function for the spinner's values.

(b) What does your intuition tell you about the value of the mean? Check your guess by evaluating an integral.

Problem 6 (Central Limit Theorem)**.** One hundred 6-sided dice are rolled, and their numbers are added up. Use the central limit theorem to estimate the probability that the sum of the numbers on the dice is between 330 and 380.

Problem 7. The Hydrogen atom is composed of one proton in the nucleous and one electron, which moves about the nucleus. In the quantum theory of atomic structure, it is assumed that the electron does not move in a well-defined orbit. Instead, it occupies a state known as an *orbital* which may be thought of as a "cloud" of negative charge surrounding the nucleous. At the state of lowest energy, called the *ground state*, the shape of this cloud is assumed to be a sphere centered at the nucleus. This sphere is described in terms of the probability density function

$$f(r) = \frac{4}{a_0^3} r^2 e^{-2r/a_0},$$

where a_0 is a physical constant known as the *Bohr radius*, and $a_0 \approx 5.29 \times 10^{-11}$ `meters`. For each $r > 0$, let $P(r)$ denote the probability that the electron will be found within the sphere of radius r meters centered at the nucleus. Then

$$P(r) = \int_0^r f(s)ds.$$

(a) Verify that f is a probability density function.

(b) Find $\lim_{r\to\infty} f(r)$.

(c) For what value of r does $f(r)$ achieve its maximum value?

(d) Graph the density function. (I recommend using Desmos.)

(e) Find the probability that an electron will be within the sphere of radius $4a_0$.

(f) At a moment in time, the distance of the electron from the nucleus is a random variable, which we will call R. A good "best guess" for this distance is the expected value. Calculate the expected distance of the electron from the nucleus in the ground state of the hydrogen atom.

(g) The variance of R expresses the *uncertainty* attached to our expectation of the electron's position. Compute $\text{Var}(R)$.

(h) The momentum M of the electron is another random variable whose density function is related to that of R (in a manner which is beyond the scope of this class). The *Heisenberg uncertainty principle* says that there is a constant $c > 0$ such that

$$\text{Var}(R)\text{Var}(M) \geq c$$

In one or two sentences, interpret this inequality result.

Problem 8. The new Pokemon TCG expansion set *Perfect Order* recently came out on Friday, March 27. The *Perfect Order* expansion set consists of 124 distinct cards. Cards can be purchased in packs consisting of 10 random cards. Assume that all cards are equally likely to appear in a pack. How many packs do you need to buy to ensure that the probability that you have purchased all 124 cards is at least 95%? Of course, the packs are \$9.00 each, and we want to spend as little money as possible, so find the *smallest* number of packs that you need to buy.

Problem 9. We are interested in testing whether a coin is balanced based on the number of heads Y in 36 flips; that is, $H_0 : p = .5$ vs $H_a : p \neq 5$. Suppose that we decide ahead of time to rejecct H_0 if $|Y - 18| \geq 4$.

(a) What is the value of α?

(b) What is the value of β if $p = .7$?

Problem 10. Please answer "true" or "false" to the following questions.

(a) The level of the test computed in Problem 9(a) is the probability that H_0 is true.

(b) The value of β in Problem 9(b) is the probability that H_a is true.

(c) In Problem 9(b), β was computed assuming the null hypothesis was false.

(d) If β was computed when $p = .55$, the value would be larger than the value of β obtained in Problem 9(b).

(e) The probability that the test mistakenly rejects H_0 is β.